

The Future of Logistics: How Autonomous Drones Could Transform India@100

As India builds toward its centenary, the biggest gains in logistics will not come from wider highways alone — they will come from the sky. From sustaining soldiers at high-altitude posts to supplying offshore platforms and remote districts, heavy-lift autonomous aircraft can move what matters, where roads cannot — on indigenous technology.



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Every developed economy has been built on a logistics revolution — railways in the nineteenth century, highways and container shipping in the twentieth, e-commerce fulfilment in the last two decades. As India charts its course to 2047, the question is whether another logistics revolution will be built here or imported.

India spends an estimated 13–14 per cent of GDP on logistics, against 8–9 per cent in advanced economies. The National Logistics Policy and PM Gati Shakti target this gap. But better roads and faster trains cannot solve geography. A third of our population lives where the last hundred kilometres are hardest — Himalayan valleys, the Northeast, island territories, riverine Bengal and tribal belts. Here, the cost of moving a kilogram of cargo decides whether produce reaches a market, vaccines reach clinics, or factories can locate outside metros.

The middle mile is the missing piece

The global drone conversation has focused on the last mile — small quadcopters delivering parcels to doorsteps. That is a real market, but it is not where Indian logistics economics will be transformed. The opportunity is the middle mile: 50-to-300-kilometre legs connecting district warehouses to taluka distribution points, ports to hinterland factories, and forward supply bases to remote communities.

Today that middle mile runs on trucks over seasonal, congested or absent roads. An autonomous vertical take-off and landing (VTOL) aircraft carrying a few hundred kilograms changes the equation. It needs no runway, road or pilot. It flies point to point at

aircraft speeds, lands on a helipad-sized clearing and repeats the cycle around the clock. Routes taking a truck eight hours through mountain switchbacks become forty-minute flights.

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At Telebortix, this is the problem we have organised the company around. Our BA-75 heavy-lift VTOL platform, progressing toward prototype flight testing, is engineered for this middle-mile envelope — meaningful payloads over long ranges, from unprepared sites, in Indian operating conditions.

Forged in the harshest terrain: the defence vertical

The most demanding logistics problem in India is sustaining our armed forces at high altitude. Thousands of soldiers hold posts along northern borders where mountain passes close for months each winter. Rations, ammunition, fuel and medical supplies move by road convoy, animal transport, porters or scarce helicopter sorties. Autonomous heavy-lift aircraft, engineered for thin air and extreme cold, enable routine unmanned resupply around the clock, without exposing aircrew or convoys to terrain and weather risk. The same capability answers civilian emergencies — cloudbursts, landslides and floods that sever road access, where the first seventy-two hours decide outcomes.

This is Telebortix’s founding vertical and where the technology is being proven. The company runs active development programmes with premier Indian Army establishments under the defence innovation framework, spanning seven platforms from nano and tactical systems to heavy-lift logistics aircraft, with field trials through 2026. High-altitude sustainment, forward resupply of ammunition and medical stores, and disaster relief are design requirements against which our aircraft are built and evaluated. A platform that earns its wings at altitude in winter will find every commercial mission comparatively forgiving.

“A platform that earns its wings supplying a post at altitude in winter will find every commercial mission comparatively forgiving.”

Autonomy is the real product

The airframe is only the visible half of the story.

The harder half — and the half that will decide which nations lead this industry — is autonomy. An aircraft requiring a skilled pilot for every flight cannot scale to thousands of daily cargo movements. One that navigates, decides and lands on its own can.

This is where India must be uncompromising about technology sovereignty. Autonomy stacks, navigation systems and flight-critical software come with strings attached when imported — export controls, denial regimes and dependencies that can be switched off in a crisis. Telebortix has built its core stack indigenously: flight autonomy, navigation for degraded or denied satellite signals, and multi-vehicle coordination software, developed in India by Indian engineers, free of foreign licensing encumbrances. What began as a defence requirement is now our commercial advantage — logistics operators, like armed forces, want systems that cannot be held hostage to someone else’s supply chain.

The dual-use dividend

There is a well-worn pattern in aerospace: technologies matured for defence become the backbone of civilian industry. Jet engines, satellite navigation and the internet travelled this road. India’s drone sector is at that inflection point now. The rigour military programmes demand — reliability in harsh environments, resilience to interference and operation without ground infrastructure — is exactly what commercial cargo aviation at scale will require.

Telebortix works across this dual-use spectrum deliberately. Our defence engagements harden the technology; our commercial roadmap gives it volume. Volume, in turn, drives down cost per flight hour — and cost per kilogram-kilometre is the metric that ultimately matters in logistics.

The commercial map extends beyond the middle mile. Offshore oil and gas is a natural first vertical: platforms and rigs in Bombay High and the KG Basin depend on helicopters and supply vessels for urgent spares, samples and consumables — missions an autonomous VTOL can fly at a fraction of the cost, in weather windows helicopters decline, without putting a crew over open water. High-value logistics is another: pharmaceutical cold chains, precision electronics, gems and time-